Literature Survey:

Project Title: Intelligent Trip Planning for Battery Electric Vehicles Using Real-Time Map Data

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1. Introduction

The rapid adoption of electric vehicles (EVs) is a key component of sustainable transportation strategies worldwide. However, limited driving range and insufficient charging infrastructure continue to create "range anxiety" among EV users—the fear that the battery will deplete before reaching the destination. Accurate prediction of energy consumption for a specific trip is essential to mitigate this anxiety and enable confident EV adoption.

Traditional vehicle simulations rely on standardized laboratory drive cycles (e.g., FTP-75, WLTP), which do not capture real-world route conditions such as elevation changes, traffic congestion, road type, and driver behaviour. Consequently, energy estimates based on these cycles may be unrealistic for actual trips. This project addresses this gap by developing a simulation-based workflow that uses real-time map data to construct realistic driving cycles and evaluate trip energy consumption for battery electric vehicles (BEVs) using MATLAB and Simulink.

2. Background and Related Work

2.1 Route-Aware EV Energy Prediction

Calvo Herrando and Borregón Sotillo (2013) developed an improved route planner-simulator that incorporates battery performance considerations for electric vehicles. Their system uses Google Maps API to obtain route geometry, distance, duration, velocity profiles, and elevation data. This information is then processed by a MATLAB-based mathematical model that simulates battery behaviour and predicts State of Charge (SOC) evolution along the route. The planner displays multiple route options, indicates trip feasibility based on current SOC, and suggests recharging stops when necessary. A real-time estimator compares predicted SOC with actual SOC during the trip to warn drivers of deviations.

This work demonstrates that integrating map-derived route data with battery models can significantly reduce range anxiety by providing drivers with clear, route-specific energy predictions.

2.2 MATLAB/Simulink-Based BEV Modelling

MathWorks provides the Virtual Vehicle Composer app, which enables users to configure, build, and analyze custom BEV models in Simulink. The app supports selection of vehicle class, powertrain architecture (including battery-electric), and vehicle dynamics, then generates a Simulink model suitable for virtual testing with drive cycles and scenarios. Logging capabilities include energy-related quantities, SOC, and vehicle position/velocity, which are directly applicable to trip-energy analysis.
Additional MathWorks resources provide reference applications and examples for BEV modelling, power and energy analysis, and component-level efficiency evaluation. These demonstrate that Simulink can model battery, motor, inverter, vehicle body, and regenerative braking systems, and compute energy flows and efficiency metrics under various operating conditions.

2.3 Map Data Integration for Driving Cycles

Google Maps Platform provides routing services through the Routes API, which returns ideal routes, distances, travel times, and traffic-aware polylines for multiple travel modes. The API supports eco-friendly route options that estimate the most fuel- or energy-efficient route based on vehicle engine type, and provides traffic conditions along the route. This enables acquisition of realistic route data, including elevation and traffic, for use in EV energy simulations.
A MATLAB Central File Exchange project demonstrates interfacing MATLAB with Google Maps APIs (Directions, Elevation) to compute time-series speed and grade profiles suitable for driving-cycle construction. This validates the feasibility of generating route-based driving data in MATLAB for simulation purposes.

3. Gap Identification and Project Contribution

Existing research and tools demonstrate that:

- Route planners can integrate map data with battery models to predict SOC and suggest charging stops.
- MATLAB/Simulink can model BEVs and analyze energy consumption under different drive cycles.
- Google Maps APIs can provide realistic route, elevation, and traffic data for use in simulations.

However, many student projects still rely on generic drive cycles or simplified route assumptions instead of fully integrating real-time map data into a custom BEV simulation workflow. The MathWorks Challenge Project “Intelligent Trip Planning for Battery Electric Vehicles Using Real-Time Map Data” explicitly calls for a simulation-based workflow that constructs realistic driving cycles from map data and evaluates trip energy using a custom BEV model in Simulink.

This project directly addresses this gap by developing an integrated workflow that:

1. Acquires real-time route and elevation data using Google Maps APIs.
2. Processes route geometry to extract distance, grade, speed limits, and traffic conditions.
3. Constructs time-speed driving cycles formatted for Simulink’s Drive Cycle Source block.
4. Builds a custom BEV model using Virtual Vehicle Composer with realistic vehicle parameters.
5. Simulates trip scenarios to evaluate SOC, energy consumption, and operational cost.
6. Visualizes route, elevation, and performance metrics using Mapping Toolbox.
7. Implements driving-mode optimization (Eco/Normal/Performance) to analyze energy-time trade-offs.

By combining route-aware planning, detailed BEV simulation, and real map data into a single coherent workflow, this project provides a practical, engineering-grade solution for intelligent EV trip planning.

4. References

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